MAST30027: Modern Applied Statistics

Assignment 4, 2026

AUTHOR
Student name and ID

Due: 5:00 pm Friday 23 October 2026

Late submissions will not be accepted unless an extension has been granted by the Special Consideration team.

This assignment is worth **10%** of your final mark.

> **How this assignment is marked**
>
> We will mark a **selected set of parts**. We will select parts worth at least **50%** of the full marks. The parts to be marked will be selected after the due date and will not be announced in advance.
>
> You must therefore attempt **every** part. A selected part that has not been attempted scores zero.
>
> To get full marks, show your working, including 1) the R commands and outputs you use, 2) the mathematical derivation, and 3) a rigorous explanation of why you reach your conclusions or answers. If you just provide final answers, you will get zero marks.

Submission and working

Create a report using the supplied QMD template and replace the author placeholder with your name and student ID. Create an RStudio Project for the assignment, keep the QMD and TXT files together in the project folder, and use relative file paths. If you include images of handwritten working, keep them in an `images` folder inside the same project.

The assignment you hand in must be typed (except for math formulas). Your answers must be clearly numbered and in the same order as the assignment questions.

Handwritten working

Math formulas may be typed using LaTeX or completed clearly by hand and included as a picture in the relevant part of the report. If you use handwritten working, scan or photograph it, crop it closely, and save it as a PNG image. The image must be upright, high contrast, legible at normal magnification, and labelled with the relevant part number.

Store the image in the `images` folder beside your QMD. Include it by placing `knitr::include_graphics()` **inside an R code chunk**, as illustrated below. Replace the example path with the relative path to your own image and remove `#| eval: false` when the file is present.

```
knitr::include_graphics("images/problem1-part-a-working.png")
```

Handwritten working must appear in the rendered report and final PDF. It may support mathematical derivations, but it does not replace required R code, output, or written explanation. Do not submit separate image files.

Final submission

Render the QMD to HTML, open the HTML in a browser, and use **Print** and **Save as PDF** to create `studentID_assignment4.pdf`. Check that all code, output, equations, figures, and handwritten working are readable and are not cut off. Submit this **single PDF** through the LMS (no other formats allowed). Do not submit the QMD, HTML, TXT, or separate image files.

It is your responsibility to upload the correct PDF on time.

The data

Three files are supplied.

File	Contents
`assignment4_prob1_2026.txt`	Final exam scores of 100 students in Modern Applied Statistics (simulated).
`assignment4_prob2_x_2026.txt`	100 observations $(x_1, \ldots, x_{100})$ for Problem 2.
`assignment4_prob2_y_2026.txt`	150 observations $(y_1, \ldots, y_{150})$ for Problem 2.

Keep all three files in the same folder as this QMD file. Remove `#| eval: false` from the data-loading chunks once the data files are correctly located and you are ready to begin.

Problem 1: Posterior and posterior predictive distributions

The file `assignment4_prob1_2026.txt` contains final exam scores of 100 students in Modern Applied Statistics. We can read the scores as follows.

```
X <- scan(file = "assignment4_prob1_2026.txt", what = double())
length(X)
mean(X)
```

You should find that `length(X)` is 100 and `mean(X)` is 75.313.

Suppose that the 100 scores are independent of each other and that they follow a Normal distribution with mean 75 and unknown precision τ. Specifically, let $x_1, \ldots, x_{100}$ be the final exam scores, and

$$x_i \sim \mathrm{N}(75, 1/\tau) \qquad \text{for } i = 1, \ldots, 100.$$

Suppose that the precision τ has a $\text{Gamma}(2, 1)$ prior distribution, where $\text{Gamma}(\alpha, \beta)$ has the pdf

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} \exp(-\beta x).$$

Part (a): Posterior distribution of the precision

Derive the posterior distribution of the precision τ conditioned on the final exam scores of 100 students, $p(\tau \mid x_1, \ldots, x_{100})$. Evaluate the parameters of the posterior distribution using the data from `assignment4_prob1_2026.txt`.

Part (b): Posterior predictive distribution

Show that the posterior predictive distribution for a new score $\tilde{x}$, $p(\tilde{x} \mid x_1, \ldots, x_{100})$, follows a three-parameter version of the t distribution (Jackman, S. (2009)), denoted by $t(\nu, a, b)$, which has the pdf

$$p(x \mid \nu, a, b) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi\nu b}} \left(1 + \frac{1}{\nu}\frac{(x-a)^2}{b}\right)^{-\frac{\nu+1}{2}}.$$

Evaluate the parameters of the posterior predictive distribution using the data from `assignment4_prob1_2026.txt`.

Problem 2: Gibbs sampling, Metropolis-Hastings and Variational Inference

Data. The files `assignment4_prob2_x_2026.txt` and `assignment4_prob2_y_2026.txt` contain 100 observations $(x_1, \ldots, x_{100})$ and 150 observations $(y_1, \ldots, y_{150})$, respectively. You can read the data as follows.

```
x <- scan(file = "assignment4_prob2_x_2026.txt", what = double(
y <- scan(file = "assignment4_prob2_y_2026.txt", what = double(
length(x)
length(y)
mean(x)
mean(y)
```

You should find that `length(x)` is 100, `length(y)` is 150, `mean(x)` is 2.728663 and `mean(y)` is -3.159025.

Model. We assume that $x_1, \ldots, x_{100}$ and $y_1, \ldots, y_{150}$ are independent and follow normal distributions:

$$x_i \sim \text{N}(\mu_1, 1^2) \qquad \text{for } i = 1, \ldots, 100,$$

$$y_j \sim \text{N}\left(\mu_2, (1/\sqrt{2})^2\right) \qquad \text{for } j = 1, \ldots, 150.$$

Prior. We impose the following bivariate normal prior for the mean parameters:

$$\begin{pmatrix}\mu_1\\ \mu_2\end{pmatrix} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad \text{with} \quad \boldsymbol{\mu} = \begin{pmatrix}0\\ 0\end{pmatrix} \quad \text{and} \quad \boldsymbol{\Sigma} = \begin{pmatrix}\frac{3}{5} & -\frac{2}{5}\\ -\frac{2}{5} & \frac{3}{5}\end{pmatrix}.$$

You can find the bivariate normal density in Lab 10.

Posterior inference using Gibbs sampling

Part (a): Conditional distributions

Derive the following conditional distributions:

$$p(\mu_1 \mid \mu_2, x_1, \ldots, x_{100}, y_1, \ldots, y_{150}) \quad \text{and} \quad p(\mu_2 \mid \mu_1, x_1, \ldots, x_{100}, y_1, \ldots, y_{150}).$$

If they are known distributions, name the distributions and specify their parameters (e.g., gamma distribution with shape $\sum_i^{100} x_i$ and scale $\sum_i^{100} x_i^2$).

Part (b): Gibbs sampling implementation

Implement Gibbs sampling to simulate samples from $p(\mu_1, \mu_2 \mid x_1, \ldots, x_{100}, y_1, \ldots, y_{150})$. Run two Gibbs sampling chains with the following two initial values.

	μ_1	μ_2
1st initial values	0	0
2nd initial values	2	-1

Please run at least 500 iterations. Make a trace plot for each of the parameters and see if samples from different chains are mixed well and behave similarly.

Part (c): Posterior summaries from Gibbs sampling

Using the simulated samples from *one* chain, for each parameter:

1. make a plot of the empirical (estimated) marginal posterior distribution,
2. estimate the marginal posterior mean, and
3. report a 90% credible interval for the marginal posterior distribution.

You can find a 90% credible interval in a number of ways. For this assignment, use 5% in each tail. Make sure to discard early iterations as a burn-in. You can decide the burn-in period from the trace plots in Part (b).

Posterior inference using the Metropolis-Hastings (MH) algorithm

Part (d): MH implementation

Implement the MH algorithm to simulate samples from $p(\mu_1, \mu_2 \mid x_1, \ldots, x_{100}, y_1, \ldots, y_{150})$. For the current values of the parameters (μ_1^c, μ_2^c), we propose new values (μ_1^n, μ_2^n) as follows:

$$\mu_1^n \sim \text{Normal}(\text{mean} = \mu_1^c, \text{variance} = 0.1^2) \quad \text{and} \quad \mu_2^n \sim \text{Normal}(\text{mean} = \mu_2^c, \text{variance} = 0.1^2).$$

Run two MH chains with the following two initial values.

	μ_1	μ_2
1st initial values	0	0
2nd initial values	2	-1

Please run at least 2000 iterations. Make a trace plot for each of the parameters and see if samples from different chains are mixed well and behave similarly.

Part (e): Posterior summaries from MH

Repeat Part (c) for the MH algorithm.

Posterior inference using Variational Inference (VI)

We will apply VI with the mean-field variational family where $q(\mu_1, \mu_2) = q_{\mu_1}(\mu_1)\, q_{\mu_2}(\mu_2)$ and use the CAVI algorithm for optimisation. The CAVI iteratively optimises each factor as follows while holding the other factors fixed:

$$q^*_{\mu_1}(\mu_1) \propto \exp\left\{E_{\mu_2}\left[\log p(\mu_1, \mu_2, x_1, \ldots, x_{100}, y_1, \ldots, y_{150})\right]\right\},$$

$$q^*_{\mu_2}(\mu_2) \propto \exp\left\{E_{\mu_1}\left[\log p(\mu_1, \mu_2, x_1, \ldots, x_{100}, y_1, \ldots, y_{150})\right]\right\},$$

where the expectations E_{μ_1} and E_{μ_2} are taken with respect to $q^*_{\mu_1}(\mu_1)$ and $q^*_{\mu_2}(\mu_2)$, respectively.

Part (f): Optimal variational factors

Derive $q^*_{\mu_1}(\mu_1)$ and $q^*_{\mu_2}(\mu_2)$ and write the corresponding distribution names and their parameters. For example, for the model and prior we considered in the lecture (slide 21 of Variational Inference), we derived and presented the following:

$q^*_\mu(\mu)$ is the pdf of $\text{N}(\mu^*, \sigma^{2*})$ and $q^*_\tau(\tau)$ is the pdf of $\text{Gamma}(a^*, b^*)$, where

$$\mu^* = \frac{\lambda_0\mu_0 + n\bar{x}}{\lambda_0 + n}, \qquad \sigma^{2*} = \frac{1}{(\lambda_0 + n)\, E_\tau[\tau]},$$

$$a^* = \frac{n+1}{2} + a_0, \qquad b^* = b_0 + \frac{\sum_i E_\mu\left[(x_i - \mu)^2\right]}{2} + \frac{\lambda_0 E_\mu\left[(\mu - \mu_0)^2\right]}{2},$$

$$E_\tau[\tau] = \frac{a^*}{b^*}, \qquad E_\mu\left[(x_i - \mu)^2\right] = \sigma^{2*} + (\mu^*)^2 - 2x_i\mu^* + x_i^2,$$

$$E_\mu\left[(\mu - \mu_0)^2\right] = \sigma^{2*} + (\mu^*)^2 - 2\mu_0\mu^* + \mu_0^2.$$

Part (g): ELBO

Derive the ELBO (up to an additive constant).

Part (h): CAVI implementation

Implement the CAVI algorithm and obtain $q^*_{\mu_1}(\mu_1)$ and $q^*_{\mu_2}(\mu_2)$ which minimise the KL divergence by applying the implemented algorithm to $x_1, \ldots, x_{100}, y_1, \ldots, y_{150}$. Set the CAVI algorithm to stop when either the number of iterations reaches 100 or the ELBO has changed by less than 10^{-5}.

Run the CAVI algorithm at least two times using different initial values and report $q^*_{\mu_1}(\mu_1)$ and $q^*_{\mu_2}(\mu_2)$ with the highest ELBO. Provide the initial values which lead to the reported $q^*_{\mu_1}(\mu_1)$ and $q^*_{\mu_2}(\mu_2)$ (such that your marker can verify your output if necessary).

For each run, check that the ELBO increases at each iteration by plotting them.

Presentation requirements

- Create your report using this QMD template and write beneath the supplied headings. You may add subheadings, but do not delete the assignment brief.
- Replace the author placeholder in the YAML with your name and student ID.
- Answers must be clearly numbered and in the same order as the assignment questions. Because the parts to be marked are not announced in advance, a missing or unlabelled answer cannot be given credit.
- To get full marks, show your working: the R commands and output you used, the mathematical derivation, and a rigorous explanation of why you reach your conclusions. Final answers with no working receive zero marks.
- All R code and results used in the analysis must be visible. Do not paste screenshots of code, console output, tables, or figures.
- The QMD must render successfully from beginning to end in a clean R session and must not rely on objects created interactively.
- Include every package-loading command required by your report. Installation commands such as `install.packages()` must not appear in the submitted report.
- Write your Gibbs, MH and CAVI implementations yourself using base R. Do not use a package that runs MCMC or variational inference for you. You may use such a package only as a check, and if you do, say so.
- Report numerical results to a sensible precision. Give every figure an informative caption, label axes clearly, and refer to each figure in the text.
- Cite any external sources, packages, or code you use.

Problem 1

Part (a): Posterior distribution of the precision

Derive $p(\tau \mid x_1, \ldots, x_{100})$, name the distribution, and evaluate its parameters using the data.

Part (b): Posterior predictive distribution

Show that $p(\tilde{x} \mid x_1, \ldots, x_{100})$ is a $t(\nu, a, b)$ distribution and evaluate ν, a and b using the data.

Problem 2

Part (a): Conditional distributions

Derive both full conditional distributions. Name each distribution and specify its parameters.

Part (b): Gibbs sampling implementation

Write your Gibbs sampler here as live R code. Run two chains from the two sets of initial values, make a trace plot for each parameter, and comment on mixing.

Part (c): Posterior summaries from Gibbs sampling

State your burn-in period. For each parameter, plot the estimated marginal posterior distribution, estimate the marginal posterior mean, and report a 90% credible interval.

Part (d): MH implementation

Write your MH algorithm here as live R code. Run two chains from the two sets of initial values, make a trace plot for each parameter, and comment on mixing.

Part (e): Posterior summaries from MH

State your burn-in period. For each parameter, plot the estimated marginal posterior distribution, estimate the marginal posterior mean, and report a 90% credible interval.

Part (f): Optimal variational factors

Derive $q^*_{\mu_1}(\mu_1)$ *and* $q^*_{\mu_2}(\mu_2)$*. Name each distribution and specify its parameters.*

Part (g): ELBO

Derive the ELBO up to an additive constant.

Part (h): CAVI implementation

Write your CAVI implementation here as live R code. Run it from at least two sets of initial values, plot the ELBO against iteration for each run, and report $q^*_{\mu_1}(\mu_1)$ *and* $q^*_{\mu_2}(\mu_2)$ *with the highest ELBO together with the initial values that produced them.*